

Security Attacks, Threats, Risks and Their Impact on Security Goals

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Course	Cryptography and Network Security
Course Code	CSA5103



What is a Security Threat?

- A **security threat** is a potential danger to a computer system or network.
- It can exploit **weaknesses** in a system.

Common security threats:



Malware



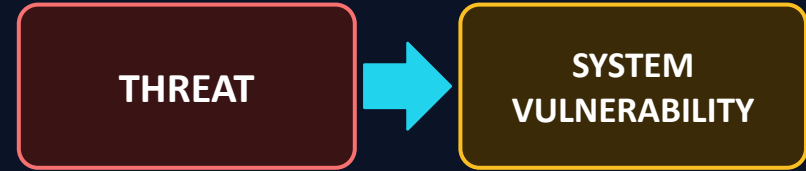
Phishing



Password attacks



Insider threats



EXAMPLE

"A hacker attempting to steal a user's password is a security threat."

What is a Security Attack?

A **security attack** is an actual attempt to access, damage, modify, or disrupt a system.



1. PASSIVE ATTACK

- Observes or monitors information
- Does not modify data
- Example: Eavesdropping

PASSIVE ATTACK



Observe Information



2. ACTIVE ATTACK

- Modifies, destroys, or disrupts information
- Example: Data modification and DoS attack

ACTIVE ATTACK



Modify / Disrupt
Information

What is Security Risk?

- **Security risk** is the possibility of loss or damage caused by a threat exploiting a vulnerability.
- A **vulnerability** is a weakness in a system.

Common vulnerabilities:



Weak passwords



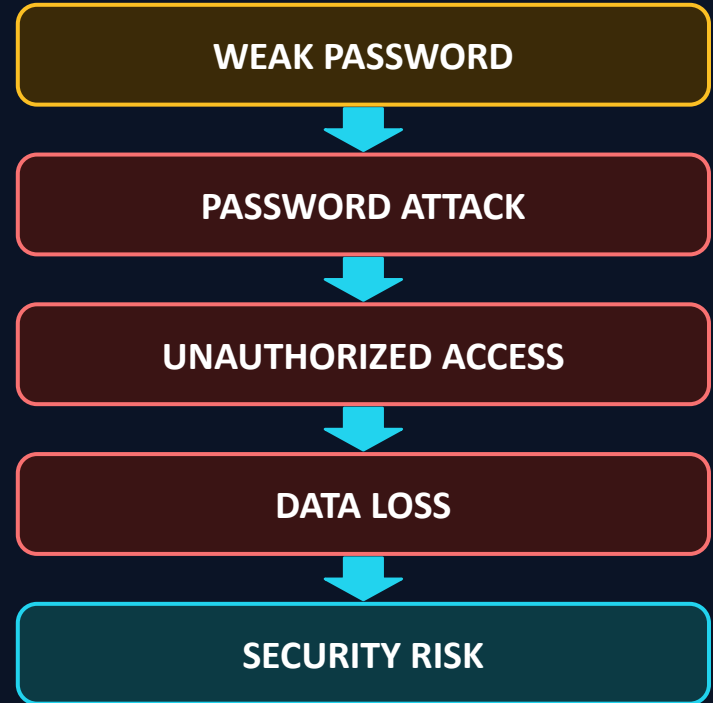
Outdated software



Poor access control



Human errors



Impact on Security Goals – CIA Triad



CONFIDENTIALITY

Prevents unauthorized access

Example: Encryption



INTEGRITY

Prevents unauthorized modification

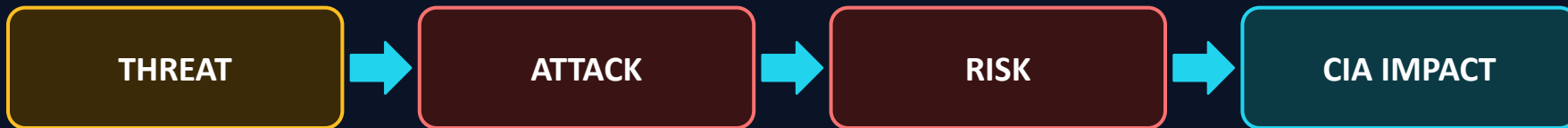
Example: Hashing



AVAILABILITY

Systems available when needed

Example: DoS protection



Conclusion

- Security threats can lead to serious cyber attacks.
- Vulnerabilities increase security risks.
- Attacks can affect Confidentiality, Integrity, and Availability.
- Strong security controls help reduce security risks.
- Continuous monitoring and security awareness are important.



“Effective cybersecurity protects Confidentiality, Integrity and Availability.”

Thank You

Steganography vs Cryptography

Hidden Security in the Digital World

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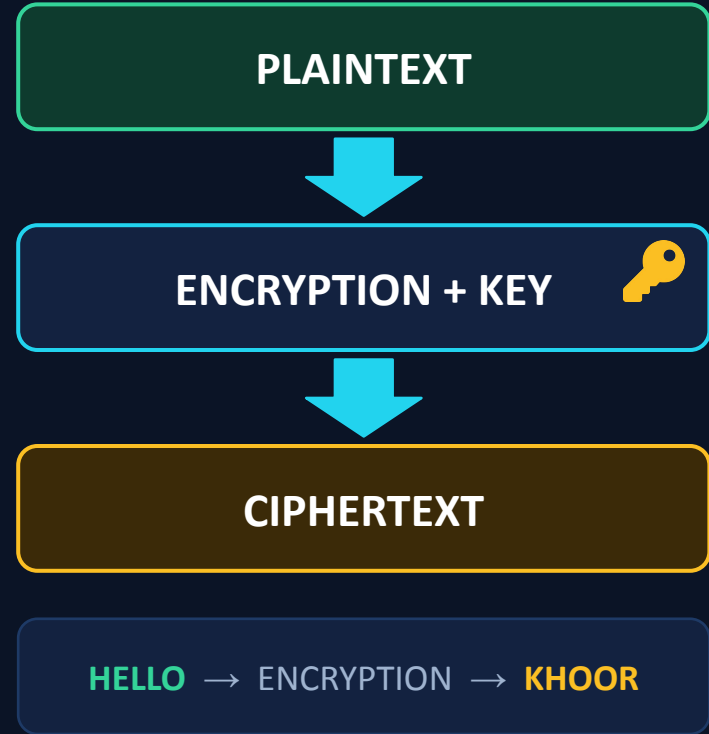
What is Cryptography?

Cryptography protects information by converting it into an unreadable form.

Original message = **Plaintext**

Encrypted message = **Ciphertext**

A **key** is used for encryption and decryption.



What is Steganography?

Steganography hides secret information inside another medium.

Secret information can be hidden inside:



Images



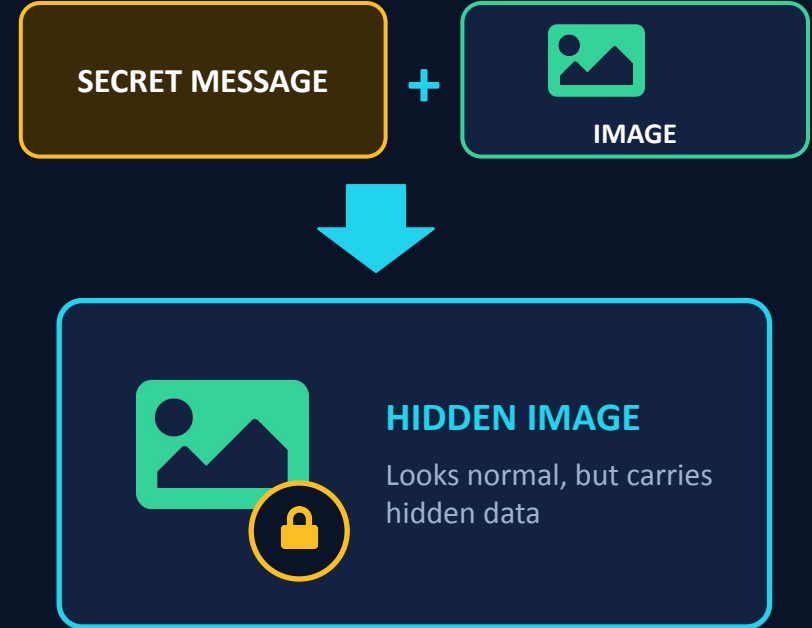
Audio



Video



Text



Steganography vs Cryptography



CRYPTOGRAPHY

- Converts data into unreadable form
- Message appears as ciphertext
- Uses encryption keys
- Protects the meaning of information
- Examples: AES, RSA

vs



STEGANOGRAPHY

- Hides data inside another medium
- Secret message is not easily visible
- Uses a carrier medium
- Hides the existence of information
- Example: Image steganography

Cryptography = Hide the Meaning

Steganography = Hide the Existence

Applications & Advantages

CRYPTOGRAPHY APPLICATIONS



Online banking



Secure communication



Password protection



Digital signatures

STEGANOGRAPHY APPLICATIONS



Secret communication



Digital watermarking



Copyright protection



Secure data transfer

ADVANTAGES



Protects sensitive information



Provides additional security



Combined with cryptography
for stronger protection

Conclusion

- Cryptography makes information unreadable to unauthorized users.
- Steganography hides the existence of secret information.
- Both techniques are useful in cybersecurity.
- Combining both can provide stronger data protection.



“Cryptography hides the message, while Steganography hides the presence of the message.”

Thank You